

VELAGAPUDI RAMAKRISHNA SIDDHARTHA ENGINEERING COLLEGE

Department of Artificial Intelligence and Data Science

RE PLASTIC INNOVATIONS

Salesforce-Based Inventory Management System

PROJECT DOCUMENTATION

STUDENT DETAILS

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1 Project Overview

1.1 Introduction

Re Plastic Innovations is a Salesforce-based inventory management solution designed to manage plastic waste, recycling centers, recycled products, customer orders, stock levels, and restocking activities in a centralized Salesforce environment. The project combines Salesforce data modelling, security configuration, formula fields, validation rules, Flow automation, Apex classes, Apex triggers, and email notification logic.

The inventory workflow is centered on recycled products. Customer orders are processed against available stock. When sufficient stock is available, the ordered quantity is deducted. When stock is insufficient, the system creates a Restock Request for the quantity required. Once the request is approved, Apex updates product stock and an email notification is generated.

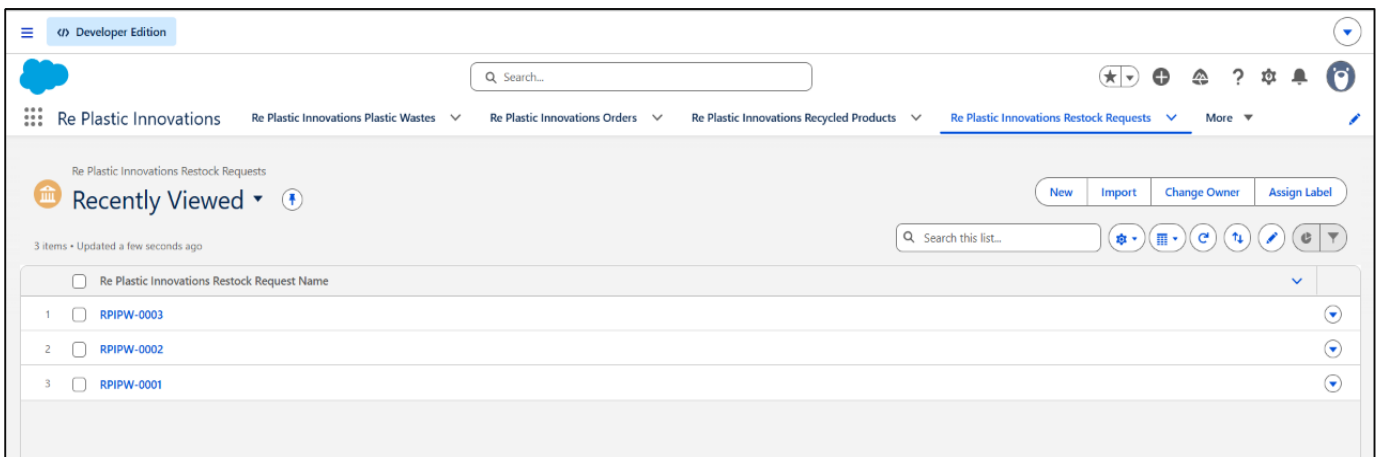
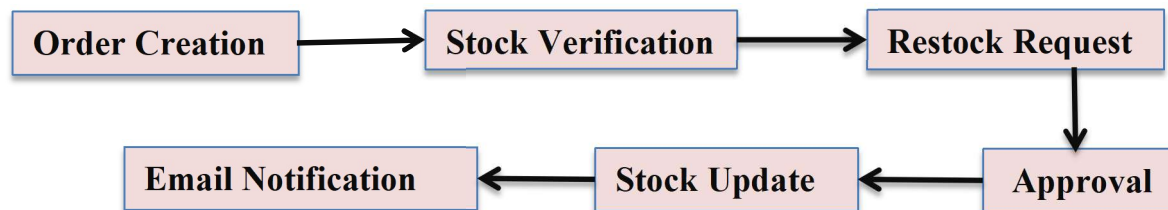


Figure 1: Re Plastic Innovations Salesforce application overview

1.2 Technology Used

The project is developed using the Salesforce Platform and Lightning Experience for managing business processes and data.

Custom Objects, Fields, Lookup Relationships, Formula Fields, and Validation Rules are used to

implement the data model and business rules.

Salesforce Flow Builder is used to automate low-stock monitoring and automatic Task creation.

Apex Classes, Triggers, and Salesforce Email Services are used for inventory processing, restock automation, and email notifications.

Table 1: Technology and Salesforce Features Used

Area	Implementation
Platform	Salesforce Platform / Lightning Experience
Data Management	Custom Objects and Custom Fields
Relationships	Lookup Relationships
Business Logic	Formula Fields and Validation Rules
Automation	Scheduled-Triggered Flow
Programming	Apex Classes and Apex Triggers
Communication	Messaging.SingleEmailMessage
Testing	Salesforce records, Apex tests and code coverage

1.3 System Overview

1. A recycled product is maintained with stock level and threshold.
2. A customer order is created for the product.
3. Apex checks available stock.
4. If sufficient stock exists, stock is reduced by the ordered quantity.
5. If stock is insufficient, a Restock Request is automatically created.
6. The request initially remains Pending.
7. When the request changes to Approved, product stock is increased.
8. An email notification is generated after approval.
9. A scheduled Flow identifies products below threshold and creates a high-priority Task.

2 Objectives and Implementation Scope

2.1 Project Objectives

- Create a structured Salesforce data model for the five custom objects.
- Maintain accurate inventory information using Salesforce fields and business rules.
- Control access using roles, profiles, users and sharing configuration.
- Prevent invalid order quantities.
- Calculate stock status using a formula field.
- Identify low-stock products automatically.
- Create Tasks for low-stock products.
- Automate stock processing using Apex.
- Create Restock Requests when stock is insufficient.

- Update stock after restock approval.
- Send email notification after approval.
- Verify the implementation through functional testing.

2.2 Implementation Scope

Table 2: Project Implementation Scope

Area	Implementation	Evidence
Data Model	Five custom objects and lookup relationships	Object Manager / Schema Builder
Security	Roles, profiles, users, sharing	Security screenshots
Business Rules	Formula and validation rules	Configuration screenshots
Flow	Scheduled low-stock automation	Flow Builder / Task
Apex	InventoryManager and two triggers	Developer Console
Email	Restock approval notification	Email screenshot
Testing	Functional and Apex testing	Test / coverage evidence

3 Requirement Analysis and Data Model

3.1 Custom Objects

Table 3: Re Plastic Innovations Custom Objects

Object	Important Fields	Purpose
Plastic Waste	Weight, Type, Collection Date, Status, Recycling Center, Location	Tracks collected plastic waste.
Recycling Center	Name, Location, Capacity	Manages recycling centers.
Recycled Product	Stock Level, Threshold, Price	Maintains recycled product inventory.
Order	Customer, Product, Quantity, Delivery Date	Stores customer orders.
Restock Request	Product, Requested Quantity, Status	Stores replenishment requests.

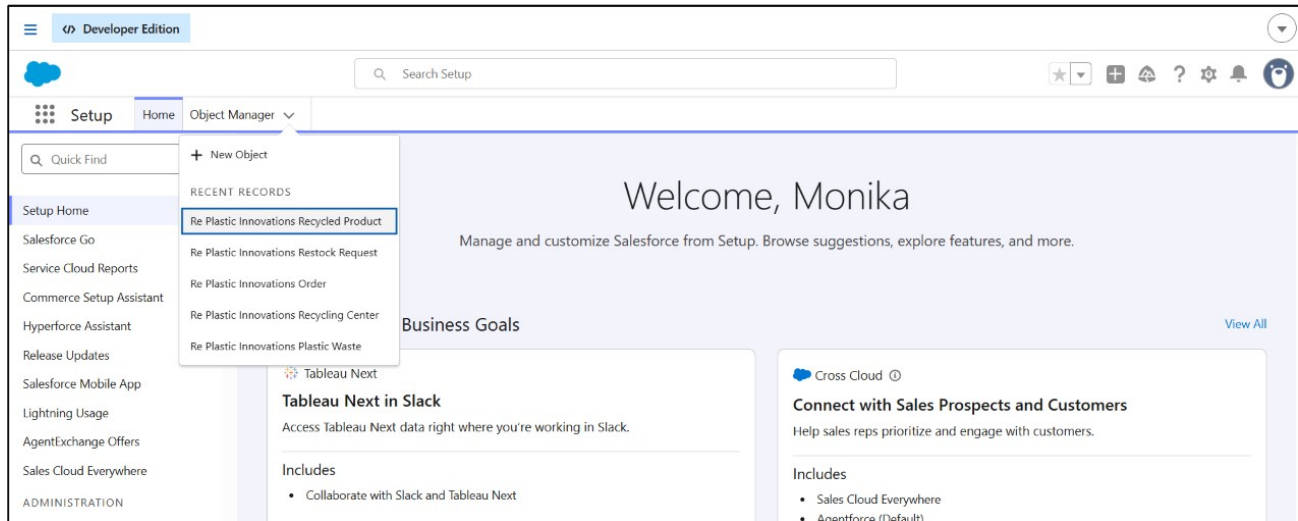


Figure 2: Custom objects configured for the application

3.2 Plastic Waste Object

Stores plastic waste collection and processing information.

3.2 Plastic Waste Object Fields

Field API Name	Data Type	Description
Name	Auto Number	Unique ID
Weight__c	Number (18,2)	Weight in kg
Type__c	Picklist	PET, HDPE, PVC, etc.
Collection_Date__c	Date	Collection date
Status__c	Picklist	Collected, Processing, Recycled
Recycling_Center__c	Lookup	Assigned center
Location__c	Geolocation	Collection location

SETUP > OBJECT MANAGER

Re Plastic Innovations Plastic Waste

Details	Fields & Relationships 10 Items, Sorted by Field Label	Q Quick Find	New
Fields & Relationships			
Page Layouts			
Lightning Record Pages			
Buttons, Links, and Actions			
Compact Layouts			
Field Sets			
Object Limits			
Record Types			
Related Lookup Filters			
Search Layouts			
List View Button Layout			
Restriction Rules			

FIELD LABEL	FIELD NAME	DATA TYPE
Collection Date	Collection_Date__c	Date
Created By	CreatedById	Lookup(User)
Last Modified By	LastModifiedById	Lookup(User)
Location	Location__c	Geolocation
Owner	OwnerId	Lookup(User,Group)
Plastic Waste Name	Name	Auto Number
Recycling Center	Recycling_Center__c	Lookup(Re Plastic Innovations Recycling Center)
Status	Status__c	Picklist
Type	Type__c	Picklist

Figure 3: Plastic Waste fields and configuration

3.3 Recycling Center Object

Represents facilities used for processing collected plastic waste.

3.3 Recycling Center Object Fields

Field API Name	Data Type	Description
Name	Text	Center name
Location__c	Geolocation	Center location
Capacity__c	Number (18,2)	Maximum capacity

SETUP > OBJECT MANAGER

Re Plastic Innovations Recycling Center

Details	Fields & Relationships 6 Items, Sorted by Field Label	Q Quick Find	New	Deleted Fields	Field
Fields & Relationships					
Page Layouts					
Lightning Record Pages					
Buttons, Links, and Actions					
Compact Layouts					
Field Sets					
Object Limits					
Record Types					

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLL
Capacity	Capacity__c	Number(16, 2)	
Created By	CreatedById	Lookup(User)	
Last Modified By	LastModifiedById	Lookup(User)	
Location	Location__c	Geolocation	
Owner	OwnerId	Lookup(User,Group)	
Re Plastic Innovations Recycling Center	Name	Text(80)	

Figure 4: Recycling Center fields and configuration

3.4 Recycled Product Object

The central inventory object. It stores product name, stock level, threshold and price.

3.4 Recycled Product Object Fields

Field API Name	Data Type	Description
Name	Text	Product name
Stock_Level__c	Number	Current stock
Threshold__c	Number	Minimum stock level
Price__c	Currency	Price per unit

SETUP > OBJECT MANAGER			
Re Plastic Innovations Recycled Product			
Details		Fields & Relationships	
Fields & Relationships		9 Items, Sorted by Field Label	
Page Layouts		Q Quick Find	
Lightning Record Pages		New Deleted Fields	
Buttons, Links, and Actions			
Compact Layouts			
Field Sets			
Object Limits			
Record Types			
Related Lookup Filters			
Search Layouts			
List View Button Layout			
Restriction Rules			
FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLL
Created By	CreatedById	Lookup(User)	
Last Modified By	LastModifiedById	Lookup(User)	
Owner	OwnerId	Lookup(User,Group)	
Price	Price__c	Currency(18, 0)	
Re Plastic Innovations Recycled Product Name	Name	Text(80)	
Stock Level	Stock_Level__c	Number(18, 0)	
Stock Level Status	Stock_Level_Status__c	Formula (Text)	
Stock Low On Product	Stock_Low_On_Product__c	Text(16)	
Threshold	Threshold__c	Number(18, 0)	

Figure 5: Recycled Product fields and configuration

3.5 Order Object

Stores customer order information and is used by the stock-processing trigger.

3.5 Order Object Fields

Field API Name	Data Type	Description
Name	Auto Number	Order ID
Customer__c	Lookup (Account)	Customer
Recycled Product__c	Lookup	Ordered product
Quantity__c	Number	Quantity ordered
Delivery Date__c	Date	Expected delivery

SETUP > OBJECT MANAGER		
Re Plastic Innovations Order		
Details	Fields & Relationships 8 Items, Sorted by Field Label Q Quick Find New Delete	
Fields & Relationships		
Page Layouts		
Lightning Record Pages		
Buttons, Links, and Actions		
Compact Layouts		
Field Sets		
Object Limits		
Record Types		
Related Lookup Filters		
Search Layouts		
List View Button Layout		
	FIELD LABEL	FIELD NAME DATA TYPE
	Created By	CreatedById Lookup(User)
	Customer	Customer__c Lookup(Account)
	Delivery Date	Delivery_Date__c Date
	Last Modified By	LastModifiedById Lookup(User)
	Owner	OwnerId Lookup(User,Group)
	Quantity	Quantity__c Number(18, 0)
	Re Plastic Innovations Order Name	Name Auto Number
	Recycled Product	Recycled_Product__c Lookup(Re Plastic Innovations Recycled Product)

Figure 6: Order fields and relationships

3.6 Restock Request Object

Represents a replenishment request generated when inventory is insufficient.

The Restock Request Object stores requests generated when the available product stock is insufficient to fulfill an order.

It records the product, requested quantity, and request status such as Pending, Approved, or Completed.

3.6 Restock Request Object Fields

Field API Name	Data Type	Description
Name	Auto Number	Request ID
Product__c	Lookup	Product to restock
Requested_Quantity__c	Number	Quantity requested
Status__c	Picklist	Pending, Approved, Completed

SETUP > OBJECT MANAGER

Re Plastic Innovations Restock Request

Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Search Layouts

Fields & Relationships
7 Items, Sorted by Field Label

Q Quick Find

New Deleted Fields

FIELD LABEL	FIELD NAME	DATA TYPE
Created By	CreatedById	Lookup(User)
Last Modified By	LastModifiedById	Lookup(User)
Owner	OwnerId	Lookup(User,Group)
Product	Product__c	Lookup(Re Plastic Innovations Recycled Product)
Re Plastic Innovations Restock Request Name	Name	Auto Number
Requested Quantity	Requested_Quantity__c	Number(18, 0)
Status	Status__c	Picklist

Figure 7: Restock Request fields and configuration

3.7 Object Relationships

Table 9: Important Object Relationships

Child	Related Object	Purpose
Plastic Waste	Recycling Center	Assigned processing center
Order	Recycled Product	Product being ordered
Restock Request	Recycled Product	Product requiring replenishment
Order	Account	Customer association

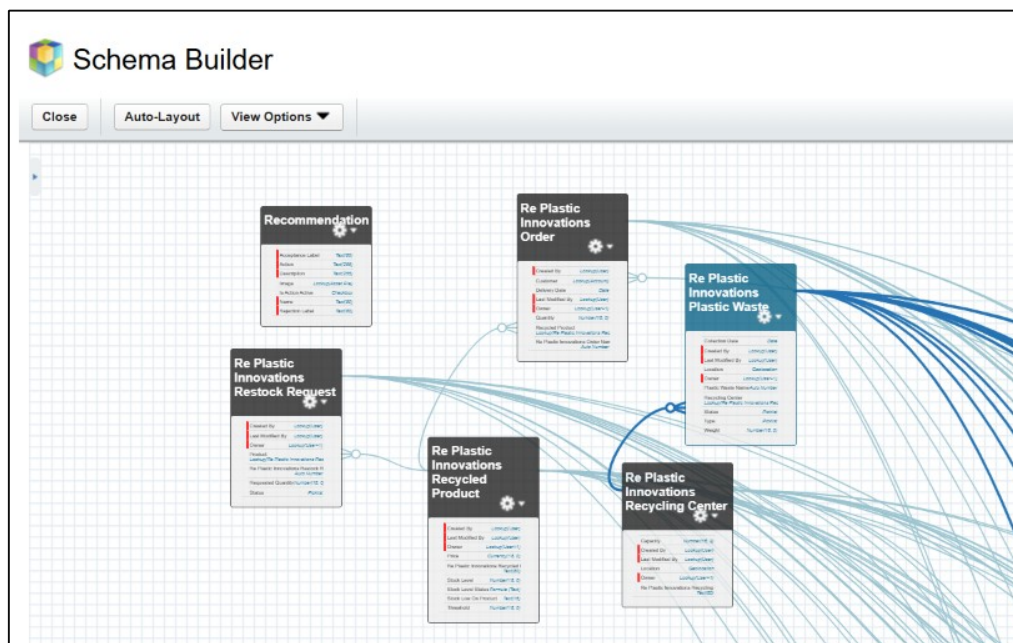


Figure 8: Schema Builder showing project data relationships

4 Salesforce Configuration and Data Security

4.1 Custom Tabs and Lightning Application

A dedicated Re Plastic Innovations Lightning application was configured with navigation items for Restock Requests, Recycling Centers, Recycled Products, Plastic Wastes and Orders.

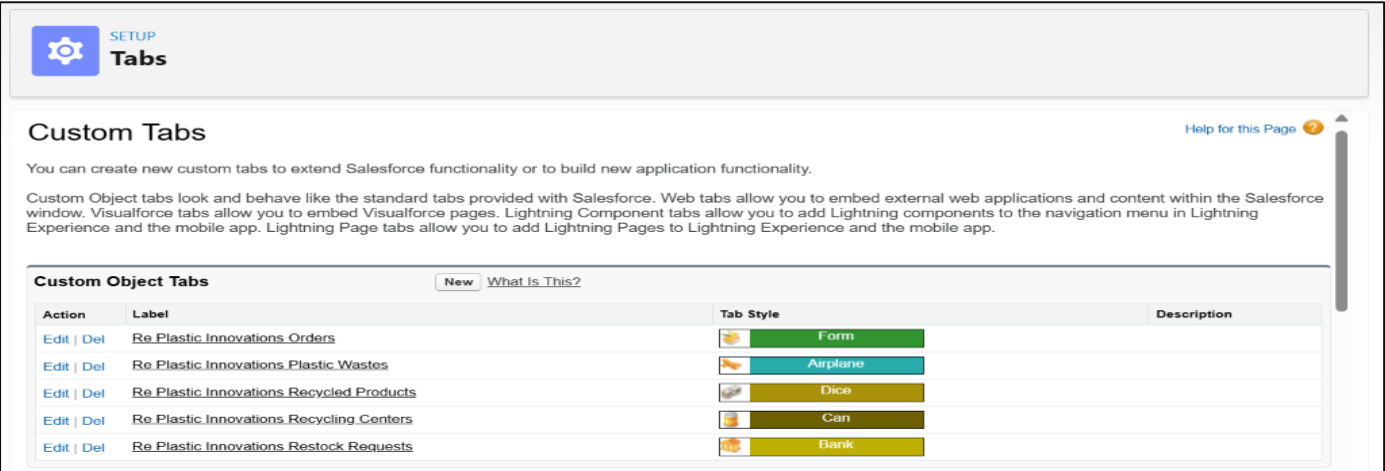


Figure 9: Re Plastic Innovations Lightning application and navigation

4.2 Role Hierarchy

The configured hierarchy places the CEO at the top, with Recycling Manager and Sales Representative reporting to the CEO, and Warehouse Supervisor reporting to the Sales Representative.

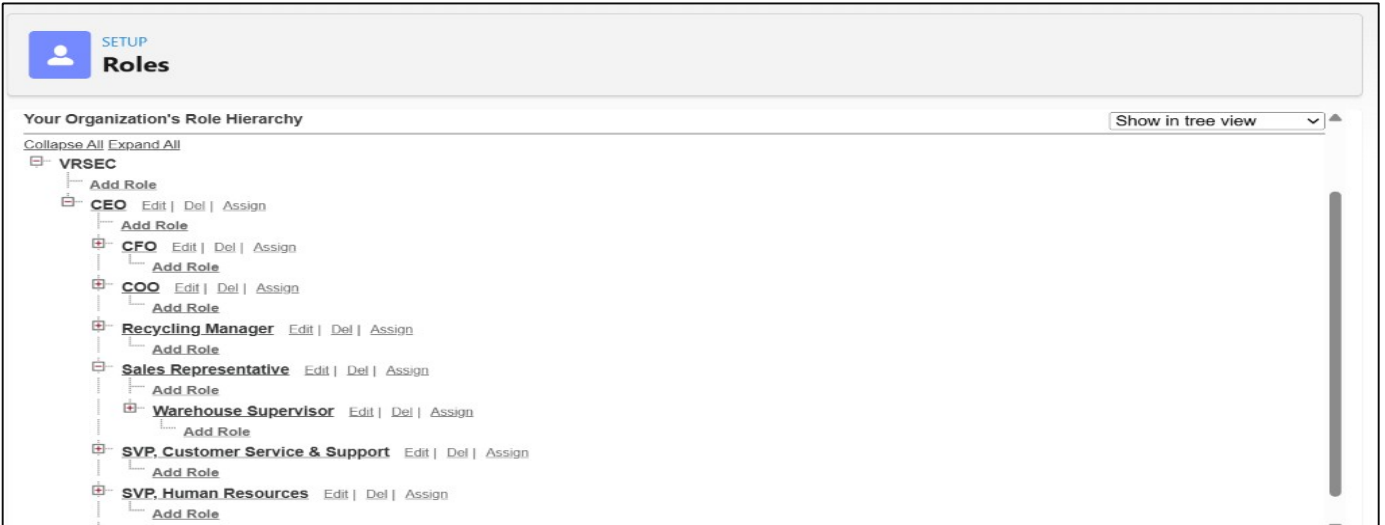


Figure 10: Salesforce role hierarchy

4.3 Profiles and Users

Table 10: Profile Configuration

Profile	Access
Platform 1	Used for Recycling Manager user

Platform 2	Read/Create Order and Account; read-only Plastic Waste and Recycled Product
Platform 3	Read/Create/Edit all custom objects

Configured users include Recycling Manager, Sales Representative and Warehouse Supervisor roles.

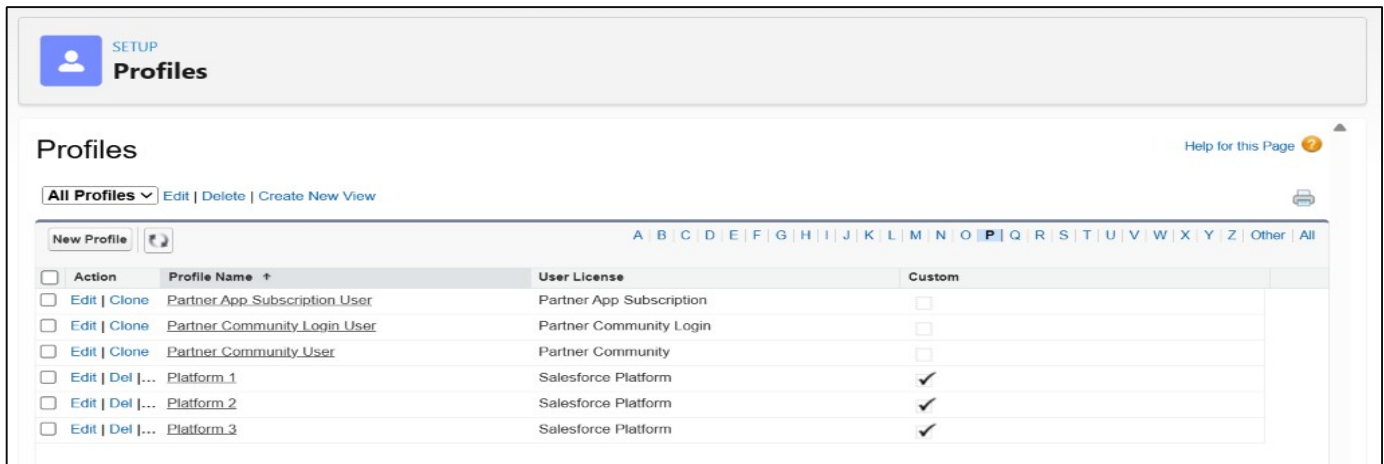


Figure 11: Profiles and project users

4.4 Record-Level Sharing

Table 11: Sharing Configuration

Rule	Object	Shared With	Access
Recycling Manager Access	Plastic Waste	Recycling Manager	Read Only
Sales Representative Access	Recycled Product	Sales Representative	Read
Warehouse Supervisor Access	Restock Request	Warehouse Supervisor	Read

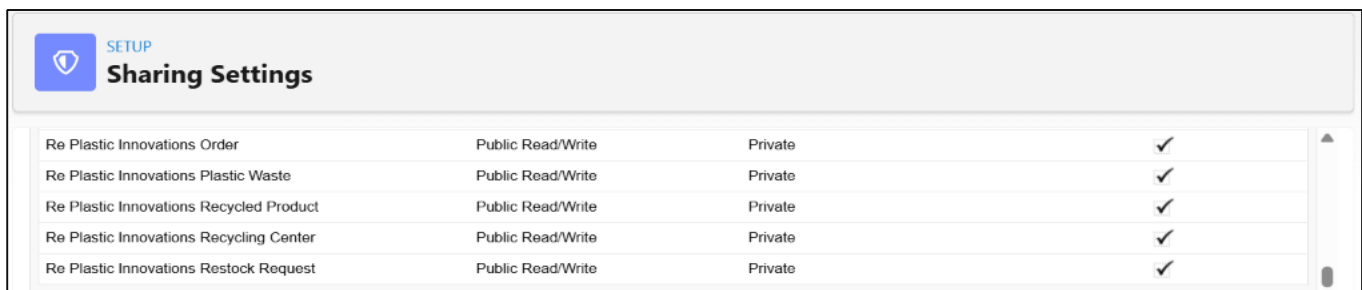


Figure 12: Salesforce sharing configuration

5 Data Quality and Business Rules

5.1 Stock Status Formula Field

A formula field named Stock Low On Product was created on Recycled Product.

IF(Stock_Level__c < Threshold__c, "Low Stock - Restock Needed", "Sufficient Stock")

Table 12: Stock Status Formula Logic

Condition	Result
Stock Level < Threshold	Low Stock - Restock Needed
Stock Level >= Threshold	Sufficient Stock

5.2 Order Quantity Validation Rule

Rule: Check_Quantity_Not_Zero

Formula: Quantity__c <= 0

Error Message: Quantity must be greater than zero.

SETUP > OBJECT MANAGER

Re Plastic Innovations Order

Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets

Re Plastic Innovations Order Validation Rule [Back to Re Plastic Innovations Order](#) [Help](#)

Validation Rule Detail [Edit](#) [Clone](#)

Rule Name	Check_Quantity_Not_Zero	Active	✓
Error Condition Formula	Quantity__c <= 0		
Error Message	Quantity must be greater than zero.	Error Location	Quantity
Description			
Created By	Monika.Ch, 9/17/2026, 12:49 AM	Modified By	Monika.Ch, 9/17/2026, 12:49 AM

[Edit](#) [Clone](#)

Figure 14: Order quantity validation rule and validation error

5.3 Collection Date Validation

Rule: Future_Date_Collection

Formula: Collection_Date__c > TODAY()

The rule prevents a future collection date from being saved.

SETUP > OBJECT MANAGER

Re Plastic Innovations Plastic Waste

Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets

Re Plastic Innovations Plastic Waste Validation Rule [Back to Re Plastic Innovations Plastic Waste](#)

Validation Rule Detail [Edit](#) [Clone](#)

Rule Name	Future_Date_Collection	Active	✓
Error Condition Formula	Collection_Date__c > TODAY()		
Error Message	Collection Date cannot be in the future.	Error Location	Collection Date
Description			
Created By	Monika.Ch, 9/17/2026, 12:58 AM	Modified By	Monika.Ch, 9/17/2026, 12:58 AM

[Edit](#) [Clone](#)

Figure 15: Collection date validation rule

6 Scheduled Flow Automation

6.1 Stock Level Is Low Flow

A Scheduled-Triggered Flow named Stock Level Is Low runs on Recycled Product and identifies products whose stock level is below the threshold.

6.2 Flow Configuration

Schedule → Decision 1 → Create Task → End

Decision: \$Record.Stock_Level__c Less Than \$Record.Threshold__c

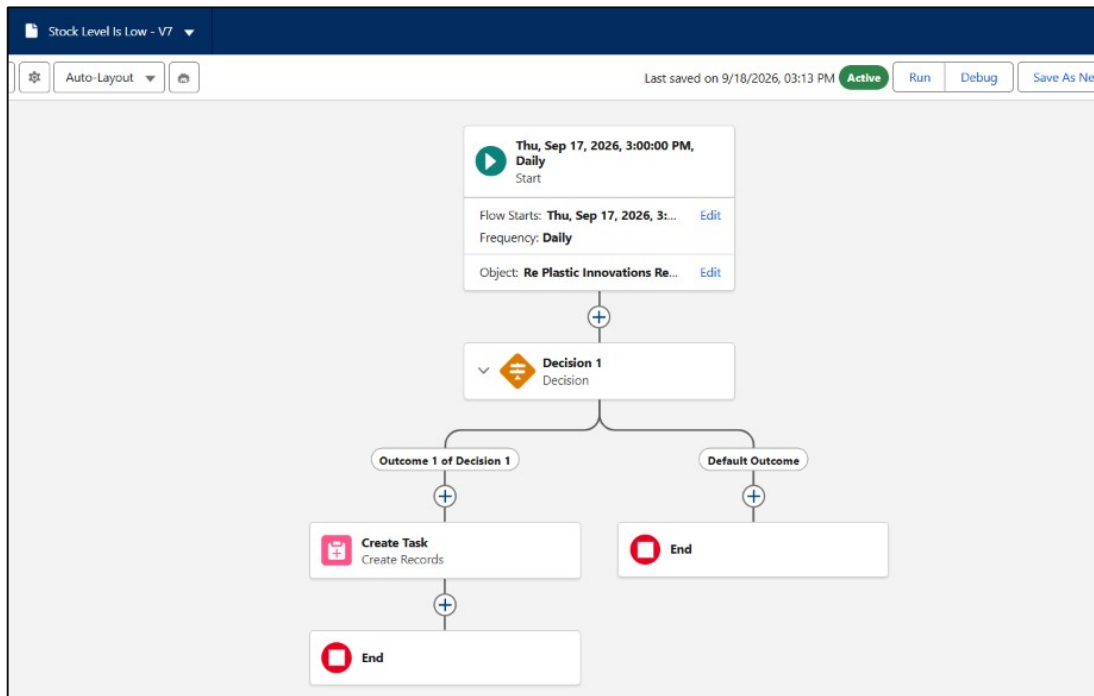


Figure 16: Stock Level Is Low Flow

6.3 Create Task Configuration

Table 13: Low Stock Task Configuration

Field	Value
Assigned To ID	Record Owner ID
Related To ID	Recycled Product Record ID
Priority	High
Subject	Please Look in this Stock Is Low Fill The Stock ASAP
Status	In Progress

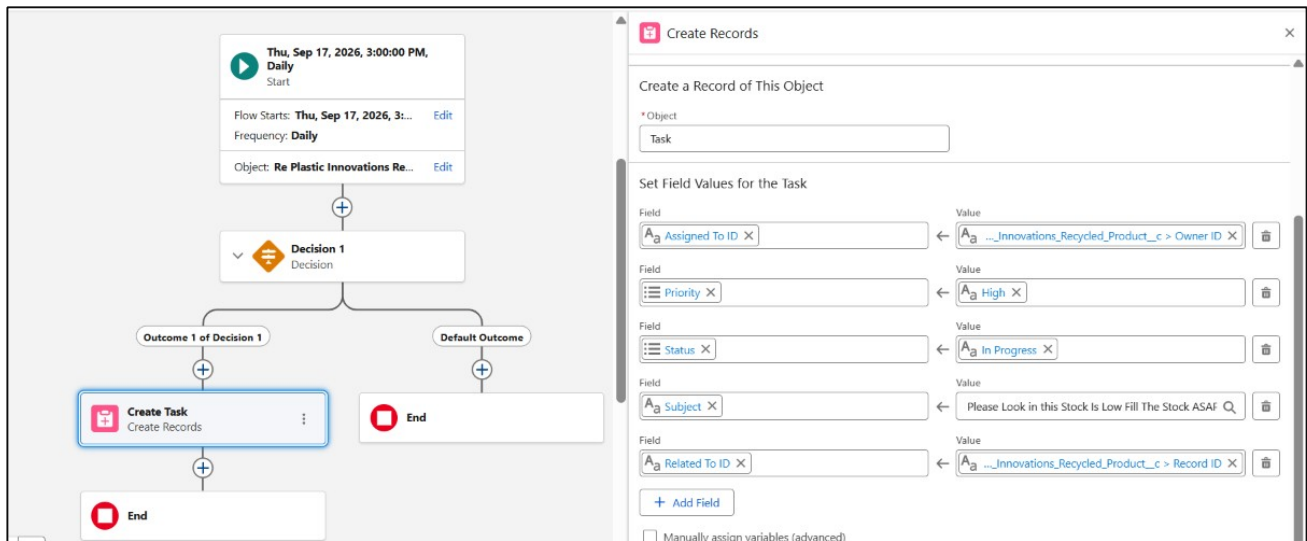


Figure 17: Create Task configuration in Flow Builder

6.4 Flow Test Result

A product with stock below its threshold is used to verify that the Flow creates the configured high-priority Task.

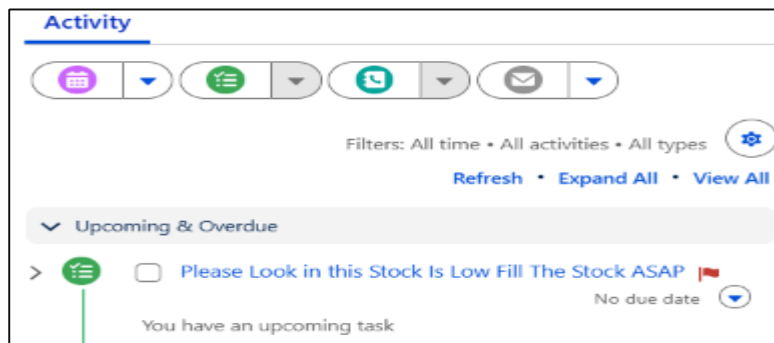


Figure 18: Automatically created low-stock Task

7 Apex Development and Trigger Automation

7.1 Apex Architecture

- InventoryManager – reusable inventory business logic.
- UpdateStockAfterOrder – processes Orders after insert.
- UpdateStockAfterRestockApproval – processes approved Restock Requests.
- EmailNotificationHelper – sends the approval email.

Open		
Entity Type	Entities	
Entity Type	Name	Namespace
Classes	DeveloperEditionUtils	devedapp
Triggers	DeveloperEditionUtilsT...	devedapp
Pages	PostInstallScript	devedapp
Page Components	PostInstallScriptTest	devedapp
Objects	InventoryManager	
Static Resources	EmailNotificationHelper	
Packages	InventoryManagerTest	

Open		
Entity Type	Entities	
Entity Type	Name	Namespace
Classes	UpdateStockAfterOrder	
Triggers	UpdateStockAfterRest...	
Pages		
Page Components		
Objects		
Static Resources		
Packages		

Figure 19: Apex classes and triggers

7.2 InventoryManager Class

Table 14: InventoryManager Responsibilities

Method	Responsibility
processOrderStock()	Checks stock; reduces stock when sufficient or creates Restock Request when insufficient.
processRestockApproval()	Adds requested quantity to product stock after approval.

The class uses Sets and Maps to collect IDs, query products and support bulk processing.

```

UpdateStockAfterRestockApproval.apxc | UpdateStockAfterOrder.apxc | InventoryManager.apxc | EmailNotificationHelper.apxc
Code Coverage: None | API Version: 67 | Go To

1 public class InventoryManager {
2     public static void processOrderStock(List<Re_Plastic_Innovations_Order__c> orderList) {
3         Set<Id> productIds = new Set<Id>();
4         for (Re_Plastic_Innovations_Order__c order : orderList) {
5             productIds.add(order.Recycled_Product__c);
6         }
7         Map<Id, Re_Plastic_Innovations_Recycled_Product__c> productMap = new Map<Id, Re_Plastic_Innovations_Recycled_Product__c>();
8         [SELECT Id, Stock_Level__c, Threshold__c FROM Re_Plastic_Innovations_Recycled_Product__c WHERE Id IN :productIds];
9         List<Re_Plastic_Innovations_Recycled_Product__c> productsToUpdate = new List<Re_Plastic_Innovations_Recycled_Product__c>();
10        List<Re_Plastic_Innovations_Restock_Request__c> restockRequests = new List<Re_Plastic_Innovations_Restock_Request__c>();
11        for (Re_Plastic_Innovations_Order__c order : orderList) {
12            Re_Plastic_Innovations_Recycled_Product__c product = productMap.get(order.Recycled_Product__c);
13            if (product != null) {
14                if (product.Stock_Level__c >= order.Quantity__c) {
15                    product.Stock_Level__c -= order.Quantity__c;
16                    productsToUpdate.add(product);
17                }
18            }
19            else {
20                restockRequests.add(new Re_Plastic_Innovations_Restock_Request__c(
21                    Product__c = product.Id,
22                    Requested_Quantity__c = order.Quantity__c - product.Stock_Level__c,
23                    Status__c = 'Pending'
24                ));
25            }
26        }
27    }
28 }

```

Figure 20: InventoryManager Apex class

7.3 Order Stock Trigger

UpdateStockAfterOrder runs after insert on Re_Plastic_Innovations_Order__c and delegates processing to InventoryManager.processOrderStock(Triiger.new).

```

1 trigger UpdateStockAfterOrder on Re_Plastic_Innovations_Order__c (after insert) {
2     if (Trigger.isAfter && Trigger.isInsert) {
3         InventoryManager.processOrderStock(Trigger.new);
4     }
5 }

```

Figure 21: UpdateStockAfterOrder Apex Trigger

7.4 Restock Approval Trigger

UpdateStockAfterRestockApproval runs after update. It detects a transition to Approved by comparing Trigger.new with Trigger.oldMap, then calls InventoryManager.processRestockApproval() and EmailNotificationHelper.sendRestockNotification().

```

1 trigger UpdateStockAfterRestockApproval on Re_Plastic_Innovations_Restock_Request__c (after update) {
2     List<Re_Plastic_Innovations_Restock_Request__c> approvedRestocks = new List<Re_Plastic_Innovations_Restock_Request__c>();
3     for (Re_Plastic_Innovations_Restock_Request__c restock : Trigger.new) {
4         if (restock.Status__c == 'Approved' &&
5             Trigger.oldMap.get(restock.Id).Status__c != 'Approved') {
6             approvedRestocks.add(restock);
7         }
8     }
9     if (!approvedRestocks.isEmpty()) {
10        InventoryManager.processRestockApproval(approvedRestocks);
11        EmailNotificationHelper.sendRestockNotification(approvedRestocks);
12    }
13 }

```

Figure 22: UpdateStockAfterRestockApproval Apex Trigger

8 Email Notification Automation

8.1 EmailNotificationHelper

The helper uses Messaging.SingleEmailMessage and Messaging.sendEmail() to notify the configured recipient after Restock Request approval.

Table 15: Email Configuration

Component	Configured Value
Subject	Restock Request Approved
Recipient	Configured email address in Apex
Body	Restock product reference and approval message

```

1 public class EmailNotificationHelper {
2
3     public static void sendRestockNotification(
4         List<Re_Plastic_Innovations_Restock_Request__c> restockRequests
5     ) {
6         List<Messaging.SingleEmailMessage> emails =
7             new List<Messaging.SingleEmailMessage>();
8
9         for (Re_Plastic_Innovations_Restock_Request__c restock : restockRequests) {
10
11             Messaging.SingleEmailMessage email =
12                 new Messaging.SingleEmailMessage();
13
14             email.setSubject('Restock Request Approved');
15
16             email.setToAddresses(
17                 new List<String>{'monika032024@gmail.com'}
18             );
19
20         }
21     }
22 }

```

Figure 23: EmailNotificationHelper Apex class

8.2 Email Notification Result

When a Restock Request changes from Pending to Approved, the trigger invokes the helper and the configured recipient receives the notification.

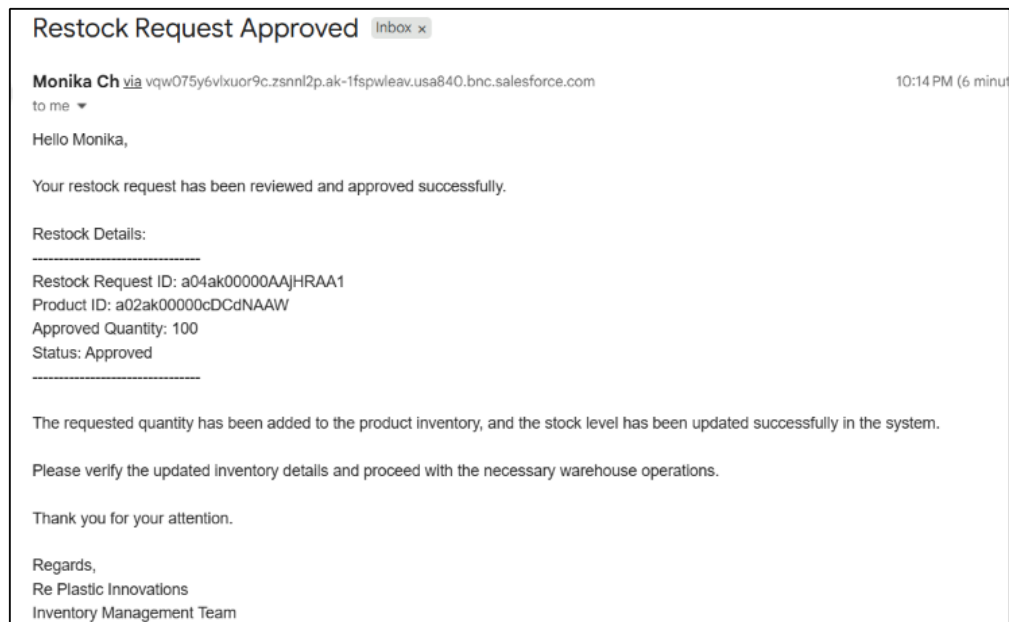


Figure 24: Restock approval email

9 End-to-End Inventory Workflow

9.1 Test Scenario

Table 16: Test Scenario

Parameter	Example
Initial Stock	50
Threshold	100

Order Quantity	150
Required Restock	100

Required Restock Quantity = 150 – 50 = 100

9.2 Step 1 – Create Order

Re Plastic Innovations Recycled Product
Recycled Plastic Pellets

Related **Details**

Re Plastic Innovations Recycled Product Name
Recycled Plastic Pellets

Owner
Monika.Ch

Stock Level
50

Threshold
100

Price
\$500

Stock Low On Product

Stock Level Status
Low Stock - Restock Needed

Created By
Monika.Ch, 9/19/2026, 9:35 AM

Last Modified By
Monika.Ch, 9/19/2026, 9:35 AM

Figure 25: Test Order created

9.3 Step 2 – Restock Request Creation

Because stock is insufficient, Apex creates a Pending Restock Request for 100.

Re Plastic Innovations Recycled Product
Recycled Plastic Pellets

Related **Details**

Re Plastic Innovations Recycled Product Name
Recycled Plastic Pellets

Owner
Monika.Ch

Stock Level
50

Threshold
100

Price
\$500

Stock Low On Product

Stock Level Status
Low Stock - Restock Needed

Created By
Monika.Ch, 9/19/2026, 9:35 AM

Last Modified By
Monika.Ch, 9/19/2026, 9:35 AM

Figure 26: Pending Restock Request

9.4 Step 3 – Approve Restock Request

The request changes from Pending to Approved and the approval trigger executes.

Re Plastic Innovations Restock Request
RPIPW-0004

Related **Details**

Re Plastic Innovations Restock Request Name
RPIPW-0004

Owner
Monika.Ch

Product
Recycled Plastic Pellets

Requested Quantity
100

Status
Approved

Created By
Monika.Ch, 9/19/2026, 9:41 AM

Last Modified By
Monika.Ch, 9/19/2026, 9:44 AM

Figure 27: Approved Restock Request

9.5 Step 4 – Product Stock Update

InventoryManager adds 100 to the original stock of 50, producing a final stock of 150.

Re Plastic Innovations Recycled Product Name
Recycled Plastic Pellets
Stock Level
150
Threshold
100
Price
\$500
Stock Low On Product

Figure 28: Updated product stock

9.6 Complete Workflow Summary

Table 17: End-to-End Workflow

Step	Action	Result
1	Create product	Product available
2	Create Order	Order inserted
3	Check stock	Insufficient stock detected
4	Create Restock Request	Requested quantity recorded
5	Keep Pending	Awaiting approval
6	Approve request	Approval trigger executes
7	Update stock	Stock increased
8	Send email	Approval notification generated

10 Functional Testing and Results

10.1 Testing Strategy

Testing covers formula fields, validation rules, scheduled Flow behavior, Task creation, order stock processing, Restock Request creation, approval processing, email notification and Apex execution.

10.2 Functional Test Cases

Table 18: Functional Test Cases

Test Case	Action	Expected Result
Quantity Validation	Enter zero/negative quantity	Record blocked and error shown
Stock Formula	Stock below threshold	Low Stock - Restock Needed shown
Low Stock Flow	Run with low stock	High-priority Task created
Sufficient Stock	Order within stock	Stock decreases
Insufficient Stock	Order exceeds stock	Restock Request created
Restock Approval	Pending → Approved	Stock increases
Email	Approve request	Email generated

10.3 Validation Rule Test Result

Inserting the corresponding Salesforce execution evidence below.

New Re Plastic Innovations Order

* = Required Information

Information

Re Plastic Innovations Order Name

Owner
Monika Ch

Customer
monika

Recycled Product
Plastic order record

Quantity
-1
Quantity must be greater than zero.

Delivery Date

We hit a snag.
Review the following fields
Quantity

Cancel Save & New Save

Figure 29: Validation rule blocking invalid Order quantity

10.4 Code Coverage

The Apex classes and triggers were tested successfully, achieving the required Salesforce code coverage for deployment.

```
orgfarm-07f11022b5-dev-ed.develop.my.salesforce.com/_ui/common/apex/debug/ApexCSIPage
File Edit Debug Test Workspace Help < >
InventoryManagerTest.apxc
Code Coverage: None API Version: 67 Run Test Go To

1 @isTest
2 public class InventoryManagerTest {
3     @testSetup
4     static void setupTestData() {
5         Re_Plastic_Innovations_Recycled_Product__c product = new Re_Plastic_Innovations_Recycled_Product__c(
6             Stock_Level__c = 50,
7             Threshold__c = 10
8         );
9         insert product;
10        List<Re_Plastic_Innovations_Order__c> orders = new List<Re_Plastic_Innovations_Order__c>{
11            new Re_Plastic_Innovations_Order__c(Recycled_Product__c = product.Id, Quantity__c = 20), // Enough stock
12            new Re_Plastic_Innovations_Order__c(Recycled_Product__c = product.Id, Quantity__c = 40) // Not enough stock
13        };
14        insert orders;
15    }
16    @isTest
17    static void testProcessOrderStock() {
18        List<Re_Plastic_Innovations_Order__c> orders = [SELECT Id, Recycled_Product__c, Quantity__c FROM Re_Plastic_Innovations_Order__c];
19        Test.startTest();
20        InventoryManager.processOrderStock(orders);
21        Test.stopTest();
22        Re_Plastic_Innovations_Recycled_Product__c product = [SELECT Stock_Level__c FROM Re_Plastic_Innovations_Recycled_Product__c];
23        System.assertEquals(10, product.Stock_Level__c, 'Stock should reduce correctly');
24        List<Re_Plastic_Innovations_Restock_Request__c> restockRequests = [SELECT Id FROM Re_Plastic_Innovations_Restock_Request__c];
```

Figure 32: Apex code coverage results

10.5 Code Coverage Results

All test cases executed successfully, confirming that the implemented Apex logic performs as expected.

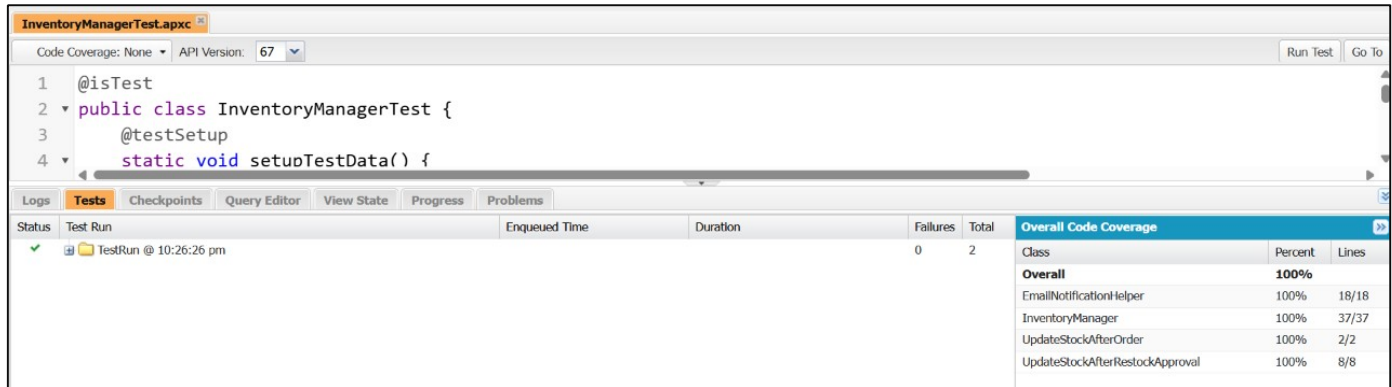


Figure 31: Apex test execution results

12 Conclusion

The Re Plastic Innovations Salesforce project provides a centralized inventory management solution using Salesforce CRM capabilities, declarative automation and Apex development. The implementation establishes a structured data model for recycled products, customer orders, restock requests, plastic waste and recycling-center records.

Formula fields and validation rules improve data quality. The Scheduled-Triggered Flow proactively identifies low-stock products and creates a high-priority Task. Apex provides transaction-level inventory automation: Orders are checked against available stock, insufficient stock generates a Pending Restock Request, and approval updates product stock and triggers an email notification.

Order → Stock Check → Restock Request → Approval → Stock Update → Email Notification